

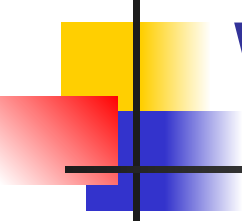
Εισαγωγή στην Κατάτμηση Εικόνας

Segmentation

- Segmentation is to partition an image into its constituent regions or objects. Segmentation should stop when the objects of interest in an application have been isolated.

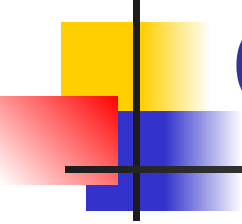


- Segmentation is an initial and vital step in pattern recognition, i.e., a series of processes aimed at overall image understanding.



Why Segment?

- Image simplification: an image may contain millions of pixels but only few regions
- Similar pixels within a region tend to belong to the same class of object
- Can provide the shape of an object
- Regions are a higher level description of image content
- To generate input for higher classification algorithms, e.g. Bayesian classifiers or Neural Networks
- Semantic description of image content allows for efficient approximate matching in Content Based Image Retrieval systems

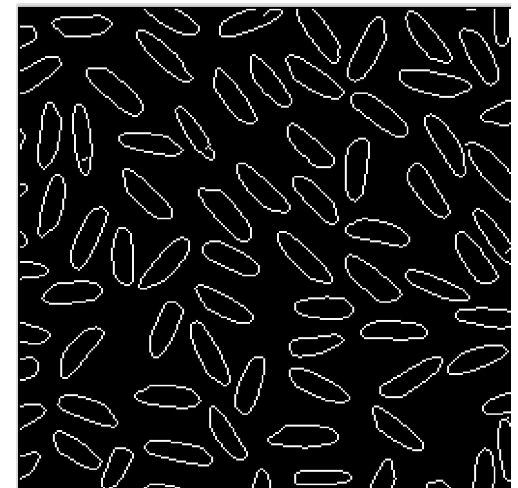
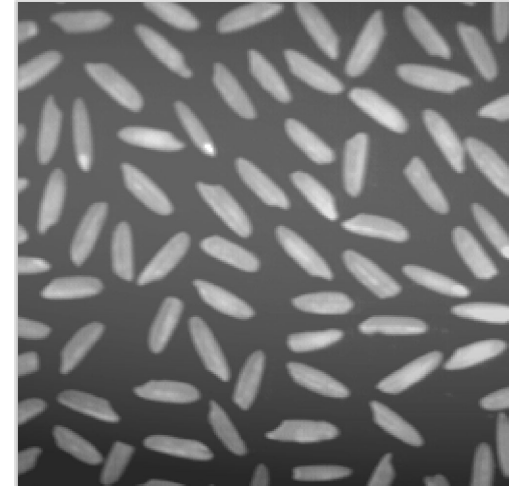


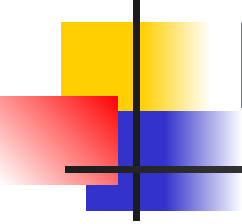
Common ways to segment

- Edge (or discontinuity) based methods:
 - Link up edges to form hopefully meaningful object boundaries
- Region based methods:
 - Region growing
 - Region splitting and merging
- Clustering methods

Edge based segmentation

- This approach identifies the strong edges in an image.
- Hopefully these correspond to object boundaries.
- Edges are extended or deleted so as to produce closed boundaries.
- Termed "Edge Relaxation".
 - takes into account not just the properties of the pixel in question but of its neighbors as well.
- Only good for simple images.
- Shape can then be used for recognition.

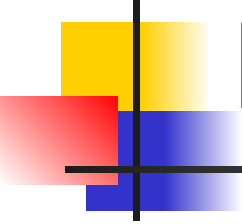




Detection of Discontinuities

- detect the three basic types of gray-level discontinuities
 - points , lines , edges
- the common way is to run a mask through the image

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9



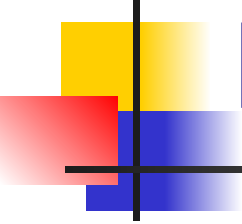
Point Detection

-1	-1	-1
-1	8	-1
-1	-1	-1

- a point has been detected at the location on which the mask is centered if

$$|R| \geq T$$

- where
 - T is a nonnegative threshold
 - R is the sum of products of the coefficients with the gray levels contained in the region encompassed by the mask.



Point Detection

- Note that the mask is the same as the Laplacian mask
- The only differences that are considered of interest are those large enough (determined by T) to be considered isolated points.

$$|R| \geq T$$

Example

-1	-1	-1
-1	8	-1
-1	-1	-1

a
b c d

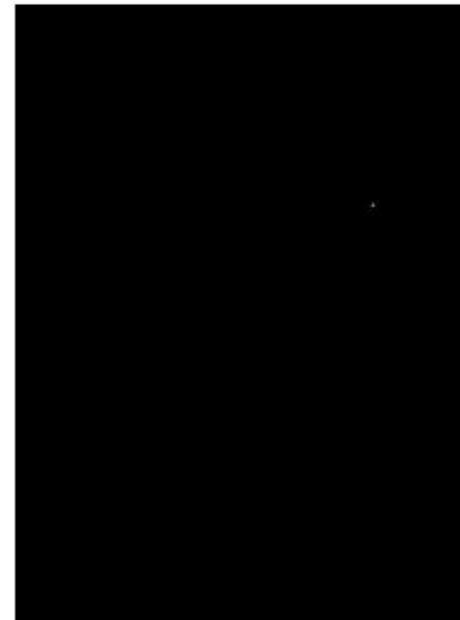
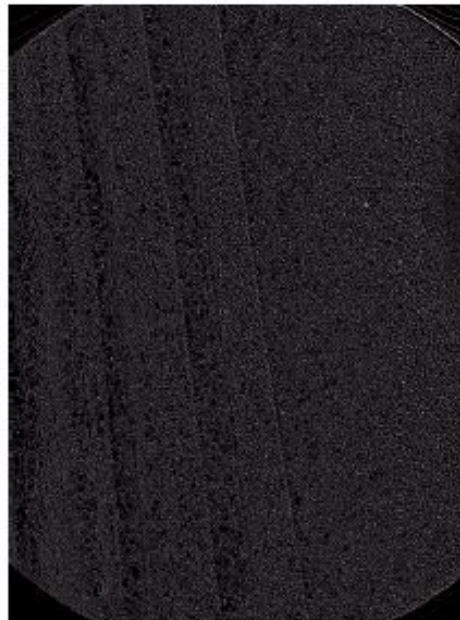
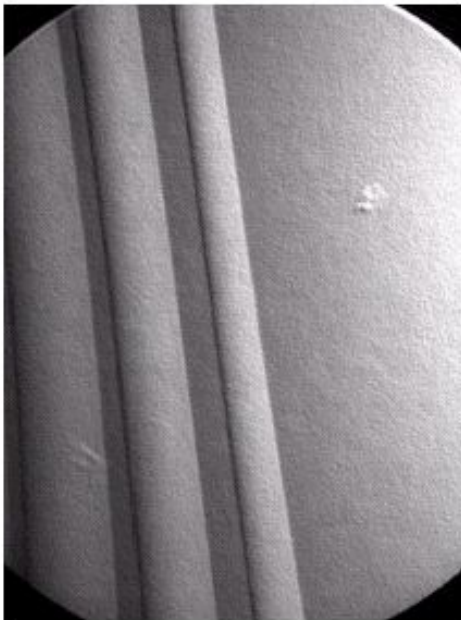
FIGURE 10.2

(a) Point detection mask.

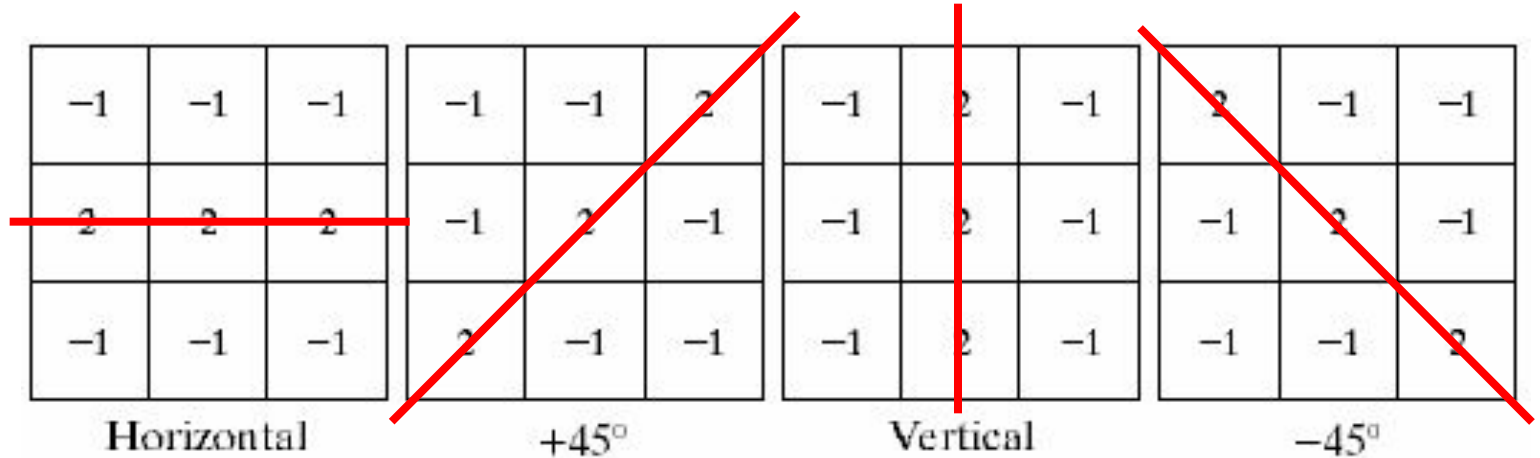
(b) X-ray image of a turbine blade with a porosity.

(c) Result of point detection.

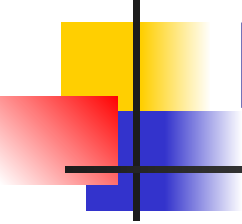
(d) Result of using Eq. (10.1-2). (Original image courtesy of X-TEK Systems Ltd.)



Line Detection



- Horizontal mask will have max response when a line of 1-pixel width passes through the middle row of the mask and when the background is constant. Similarly for the other masks.
- Note: the preferred direction of each mask is weighted with a larger coefficient (i.e., 2) than other possible directions.
- To detect dark lines we need $R < T$ and $T < 0$; to detect bright lines we set $R > T > 0$; to detect all lines, set $|R| > T > 0$.

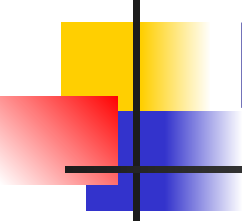


Line Detection

- Apply every mask on the image
- let R_1, R_2, R_3, R_4 denote the response of the horizontal, +45 degrees, vertical and -45 degrees masks, respectively.
- if, at a certain point in the image

$$|R_i| > |R_j|,$$

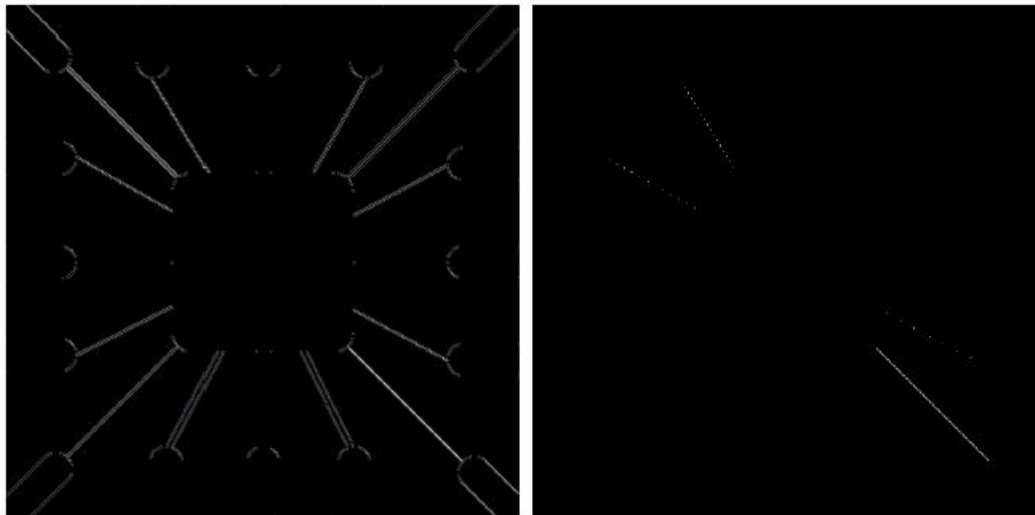
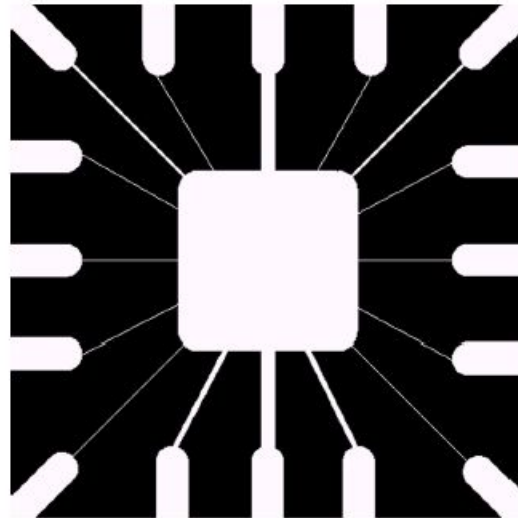
for all $j \neq i$, that point is said to be more likely associated with a line in the direction of mask i .



Line Detection

- Alternatively, if we are interested in detecting all lines in an image in the direction defined by a given mask, we simply run the mask through the image and threshold the absolute value of the result.
- The points that are left are the strongest responses, which, for lines one pixel thick, correspond closest to the direction defined by the mask.

Example



a
b c

FIGURE 10.4
Illustration of line
detection.
(a) Binary wire-
bond mask.
(b) Absolute
value of result
after processing
with -45° line
detector.
(c) Result of
thresholding
image (b).

Edge Detection

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

a
b c
d e
f g

FIGURE 10.8

A 3×3 region of an image (the z 's are gray-level values) and various masks used to compute the gradient at point labeled z_5 .

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$G = |G_x| + |G_y| \text{ or}$$

$$G = \sqrt{G_x^2 + G_y^2}$$

$$\alpha = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

Detection of diagonal edges

0	1	1	-1	-1	0
-1	0	1	-1	0	1
-1	-1	0	0	1	1

Prewitt

0	1	2	-2	-1	0
-1	0	1	-1	0	1
-2	-1	0	0	1	2

Sobel

a	b
c	d

FIGURE 10.9 Prewitt and Sobel masks for detecting diagonal edges.

Edge Detection: Example

a	b
c	d

FIGURE 10.10

(a) Original image. (b) $|G_x|$, component of the gradient in the x -direction. (c) $|G_y|$, component in the y -direction. (d) Gradient image, $|G_x| + |G_y|$.



Edge Detection: Example



a	b
c	d

FIGURE 10.11
Same sequence as in Fig. 10.10, but with the original image smoothed with a 5×5 averaging filter.

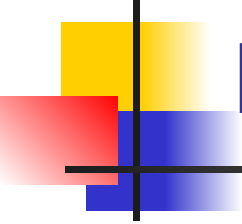
Edge Detection: Example



a b

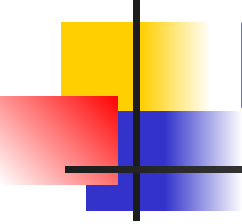
FIGURE 10.12
Diagonal edge
detection.

(a) Result of using
the mask in
Fig. 10.9(c).
(b) Result of using
the mask in
Fig. 10.9(d). The
input in both cases
was Fig. 10.11(a).



Edge Linking & Boundary Detection

- The edge detection algorithms are followed by linking procedures to assemble edge pixels into meaningful edges.
- Basic approaches
 - Local Processing
 - Global Processing via the Hough Transform



Local Processing

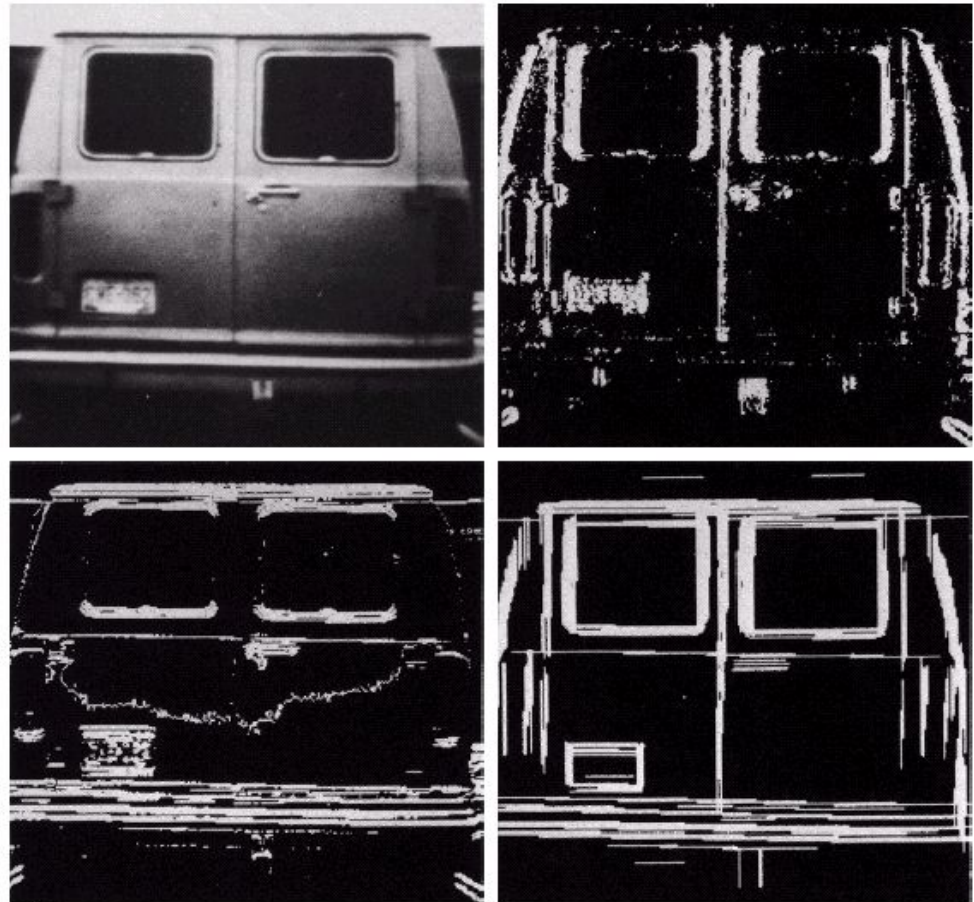
- analyze the characteristics of pixels in a small neighborhood (say, 3×3 , 5×5) around every edge pixel (x, y) in an image.
- all points that are similar according to a set of predefined criteria are linked, forming an edge of pixels that share those criteria.

Edge Linking: Local Processing

a b
c d

FIGURE 10.16

(a) Input image.
(b) G_y component
of the gradient.
(c) G_x component
of the gradient.
(d) Result of edge
linking. (Courtesy
of Perceptics
Corporation.)



$$|G(x,y) - G(x',y')| \leq T$$

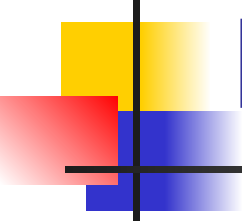
$$|a(x,y) - a(x',y')| < A$$

where

$$G(x,y) = |\nabla f(x,y)|$$

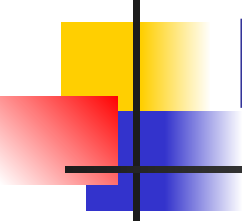
and

$$a(x,y) = \angle \nabla f(x,y)$$



k-means Clustering

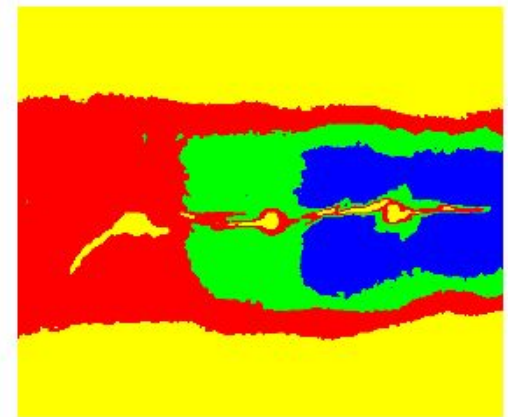
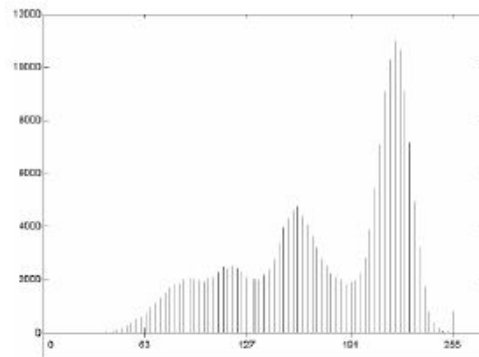
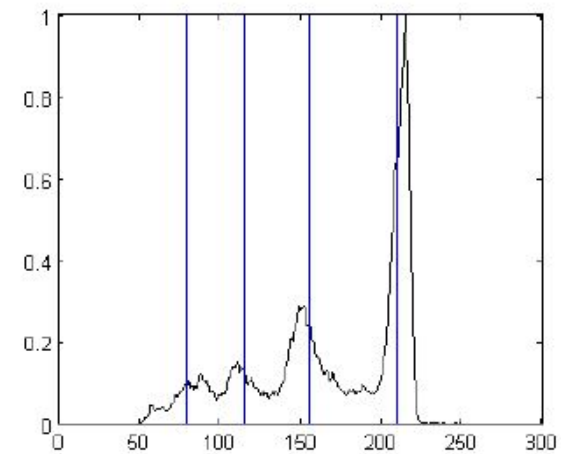
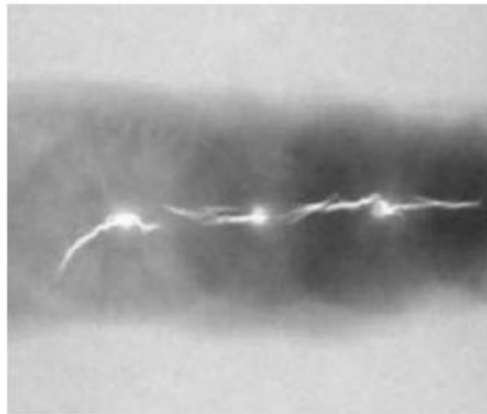
- Non-parametric, no real model - just means
- More general than above techniques, no less heuristics
- Purely interested in the global data distribution
- Want to minimize the distance between data and nearest mean
- PROBLEM: we don't know where the means are to start with – so solve iteratively
- Some prior knowledge of image statistics can be useful

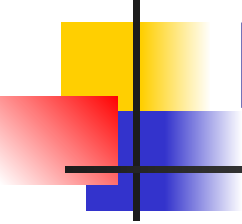


k-means Clustering

- Choose a value for k (the number of means or cluster centers)
- Make an initial distribution of these means over the histogram/data (can be random)
- Each pixel is associated with its nearest mean
- For each cluster the mean is now recalculated – this moves the means towards the true means of the data
- Repeat this process until it stabilises (converges), i.e., the means do not change anymore
- Each pixel is then labeled according to which cluster it belongs
- Final clean up may involve merging the smaller regions

k-means Clustering Example





Region Based Segmentation

- Basic Formulation

$$(a) \bigcup_{i=1}^n R_i = R$$

(b) R_i is a connected region, $i = 1, 2, \dots, n$

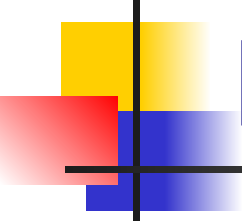
(c) $R_i \cap R_j = \emptyset$ for all i and j , $i \neq j$

(d) $P(R_i) = \text{TRUE}$ for $i = 1, 2, \dots, n$

(e) $P(R_i \cup R_j) = \text{FALSE}$ for $i \neq j$

$P(R_i)$ is a logical predicate property defined over the points in set R_i

eg. $P(R_i) = \text{TRUE}$ if all pixel in R_i have the same gray level



Region Growing

- start with a set of "**seed**" points
- growing by appending to each seed those neighbors that have similar properties such as specific ranges of gray level

select all seed points with gray level 255

Region Growing

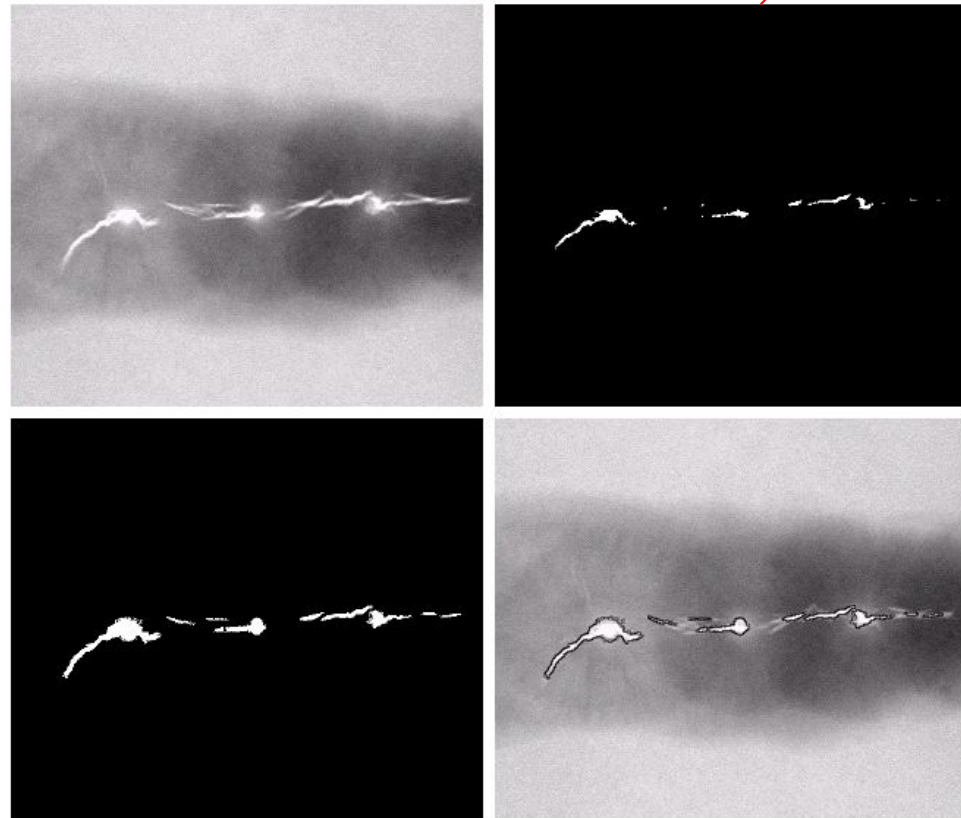
criteria:

1. the absolute gray-level difference between any pixel and the seed has to be less than 65
2. the pixel has to be 8-connected to at least one pixel in that region (if more, the regions are merged)

a b
c d

FIGURE 10.40

(a) Image showing defective welds. (b) Seed points. (c) Result of region growing. (d) Boundaries of segmented defective welds (in black). (Original image courtesy of X-TEK Systems, Ltd.).



Histogram of Fig. 10.40a

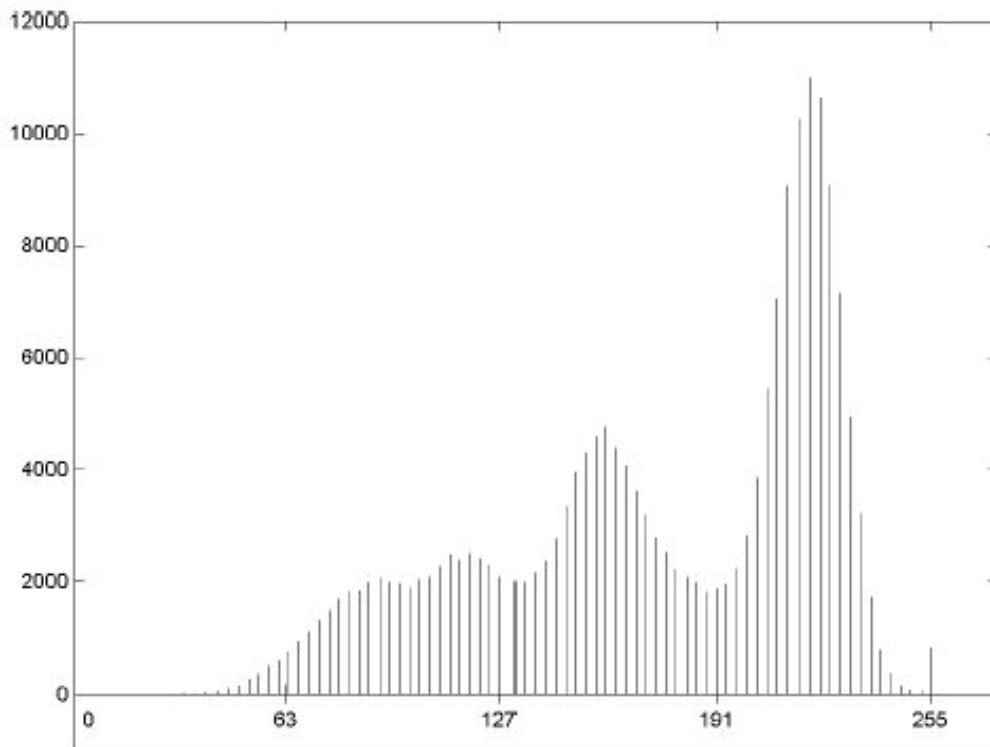


FIGURE 10.41
Histogram of
Fig. 10.40(a).

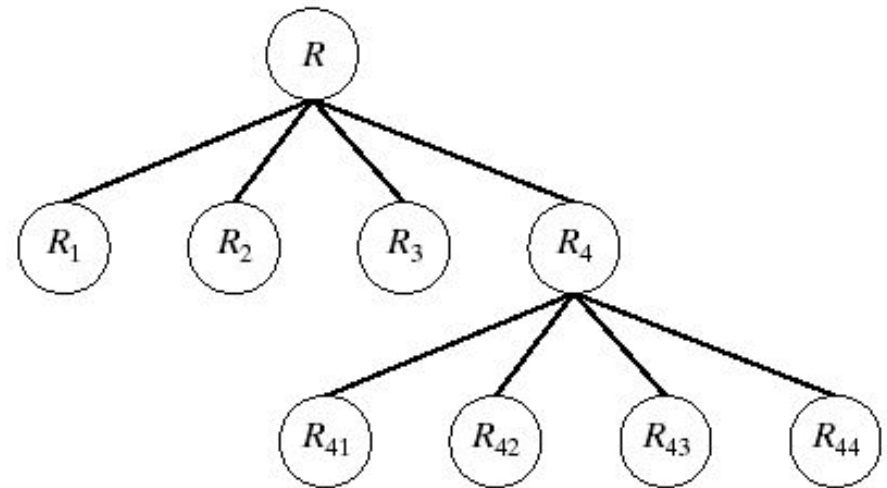
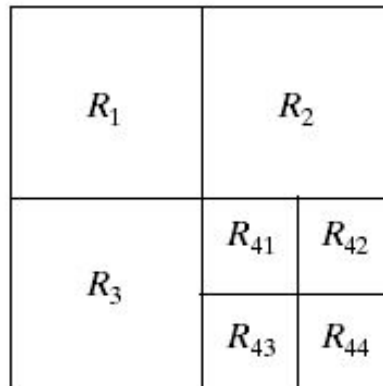
used to find the criteria of
the difference gray-level
between each pixel and
the seeds

Region splitting and merging

a b

FIGURE 10.42

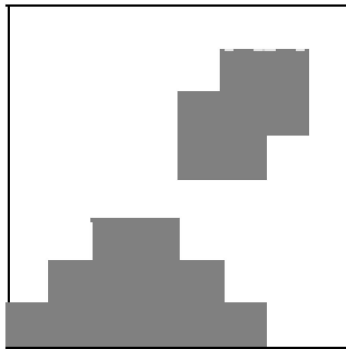
(a) Partitioned image.
(b) Corresponding quadtree.



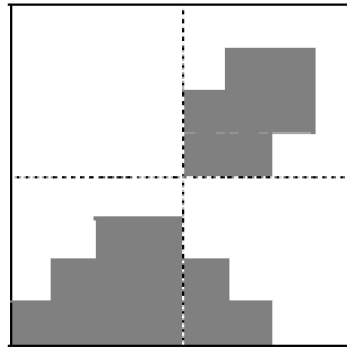
Quadtree

1. Split into 4 disjoint quadrants any region R_i for which $P(R_i) = \text{FALSE}$
2. Merge any adjacent regions R_j and R_k for which $P(R_i \cup R_k) = \text{TRUE}$
3. Stop when no further merging or splitting is possible.

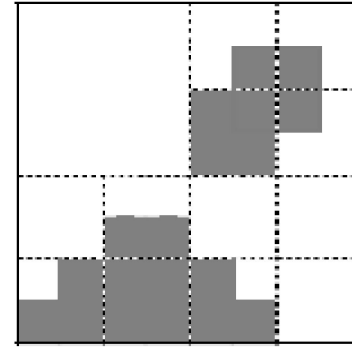
Synthetic Image Example



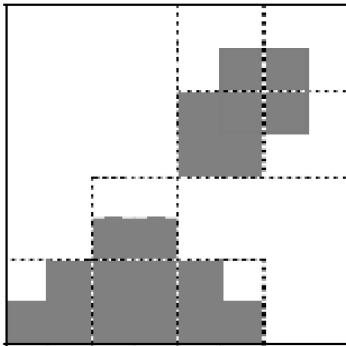
(a) Initial image



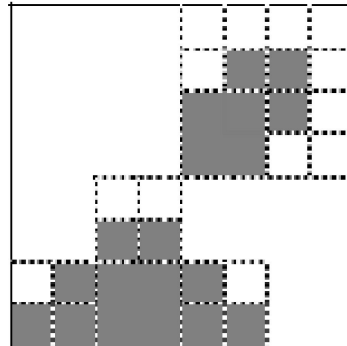
(b) Split



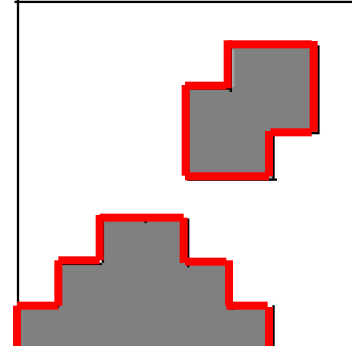
(c) Split



(d) Merge



(e) Split



(f) Merge